Journal of Nuclear Materials 573 (2023) 154108

Contents lists available at ScienceDirect

Journal of Nuclear Materials

journal homepage: www.elsevier.com/locate/jnucmat

Influence of point defect accumulation on in-pile thermal conductivity degradation: Fuel rod defect distribution and deviation between in-pile and post irradiation thermal conductivity

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ARTICLE INFO

Article history:
Received 31 March 2022
Revised 25 September 2022
Accepted 28 October 2022
Available online 30 October 2022

Keyword:
Defect distribution
Point defect characterization
Microstructure evolution
Radiation damage

ABSTRACT

As nuclear fuel burn-up increases, its thermal conductivity degrades due to the accumulation of defects that lead to increased phonon scattering rates. This results in a rise in the centerline temperature of the fuel rod, whereby heat generation must be decreased to avoid undesired behavior such as fuel melting and extensive fission gas release. Fuel performance codes are utilized to optimize the fuel's operational conditions; while they are based on established physical principles, their empirical nature limits their predictive capabilities. Recently, an effort has been made to develop predictive fuel performance codes for commonly used nuclear fuels, as well as accelerated qualification of advanced nuclear fuels. In this report, we elaborate on the importance of careful analysis of point defects' impact on thermal conductivity in fuel performance analysis. A model is presented where point defect concentration is estimated based on Rate Theory modeling and used as input to the Klemens-Callaway model to calculate their contribution to the degradation of thermal conductivity in UO_2 under prototypical irradiation conditions. This analysis suggests that point defect concentration is significant at the rim of the fuel pellet and neglecting this leads to underestimation of the centerline temperature, which may have consequences on fission gas behavior.

1. Introduction

Nuclear performance codes are one of the tools by which nuclear reactor burnup margins have steadily increased over the past three decades [1,2]. Initially what started as nuclear fuel property correlations and computer codes such as MATPRO 9 in 1976, have developed into highly cultivated programs that can be used to characterize fuel behaviors under irradiation [3,4]. Since MATPRO, engineering scale fuel performance codes (FPCs) such as FRAPCON, TRANSURANUS, and more recently BISON, have taken a more comprehensive role in the calculation of thermo-mechanical properties of nuclear fuels and their response to neutron irradiation [4–8]. As is the case with FRAPCON and TRANSURANUS, both are based on semi-empirical datasets taken from measurements of nuclear reactors in operation sometimes spanning up to a decade [9]. As meticulous as it is, FPCs can only be used for the reactor conditions that have been measured. Experimentation of test reactors has been used to bridge the gap between normal and off-normal, or transient, operating conditions. This additional experimentation can take many years to achieve extensive data about these off-normal conditions and is continuously being updated [9]. Recent efforts have focused on accelerated fuel qualification (AFQ), utilizing advanced modeling and simulation early in the qualification process to reduce the number of experiments [10,11]. Therefore, there is a need for the development of predictive models that can be used to determine fuel behavior outside of the bounded parameter space set by current experimental data [12]. One realization of this effort is MARMOT code which provides input to BISON and considers details of microstructure evolution to predict material properties [13]. While MARMOT has made significant developments in thermal conductivity predictions targeting processes at the mesoscale, notably less effort has focused on the atomic level understanding of fuel behavior [14–16].

The burnup of BWRs and PWRs has steadily increased in the U.S. over the past two decades as advancements in nuclear fuels, and FPCs allow reactor operators to extract more thermal energy from UO_2, the most common nuclear fuel in the U.S [17]. The thermal conductivity of the nuclear fuel is an important material property of interest to reactor scientists as it is crucial to the calculation of the centerline temperature of the fuel rod [18,19]. The degradation of thermal conductivity forces reactor operators to decrease the power density inside of the reactor to maintain the safety mar-

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https://doi.org/10.1016/j.jnucmat.2022.154108
0022-3115/

gin [20,21]. For this reason, much emphasis has been given to this property in FPCs, and its dynamic progression under irradiation [22].

The interaction of energetic particles with solid materials is a multiscale process. It ultimately leads to displacement damage in addition to fission and transmutation reactions. Microstructure evolution in fuel material is governed by the behavior of fission fragments [23]. Damage caused by fission fragments and heavy transuranic elements in the fuel increases defect concentration and deteriorates the fuel's physical properties [18,23,24]. These fission fragments collide with an atom of either oxygen or uranium, with high enough energies to displace the atom from its equilibrium location in the lattice and create a vacancy and interstitial (Frenkel pair). The incident and displaced particles propagate further creating a damage cascade until their energies reduce below the displacement threshold [20,25,26]. The atoms displaced from their lattice sites can migrate to form extended defects that arise from the accumulation of several point defects (PDs) [14,27-29].

The accumulation of these effects causes a reduction in thermal conductivity as a result of an increase in phonon scattering rates [12,30,31]. At high temperatures, phonons govern heat transfer in ceramics [32,33]. Crystal imperfections and phonon-phonon interactions such as Umklapp scattering limit the lattice thermal conductivity [14,34]. At high radiation doses, defect concentrations contribute to the largest reduction of thermal conductivity of the lattice, particularly near the rim of fuel rods [31,35]. The thermal conductivity of the solid is reduced until the point of the amorphous limit of thermal conductivity, where a highly disordered lattice exists and the phonon mean free path is comparable to the lattice constant [36].

Fission gas bubbles represent a common feature of radiation damage, formed from voids, and filled by inert gas fission products, that have received significant attention from a mesoscale thermal transport perspective [22,37,38]. Microstructure defects exhibit significant accumulation around the rim of fuel rods due to the lower temperature, at which defects are less mobile and recombine less [38]. The rim effect, which is characteristic of the high burnup structure (HBS), is thought to arise from the increase in burnup localized near the rim of the fuel rod, along with the accompanying increase in radiation damage [4,22,38-41]. This induces the formation of the microstructure, which is characterized by tangled dislocation loops, grain size reduction, polygonization, coarsening of pores, and inherent microstructural change that occurs above 40 MWd/kgU [35,42,43]. The overall result is a disordered lattice that decreases phonon mean free paths and greatly reduces thermal conductivity [21].

Extensive research has been devoted to understanding changing material properties, particularly thermal conductivity, detailed in a recent review [12]. Comprehensive reports by Lucuta *et. al.* and Ronchi *et. al.* presented empirical models for thermal conductivity degradation inspired mechanistic understandings of phonon-mediated thermal transport as a function of fuel burnup, based upon experimental measurements of irradiated UO_2 [32,44]. While these empirical correlations are convenient for implementation in FPC, they don't consider the detailed microstructure of the fuel, therefore limiting utilization of their correlation outside of the parameter space of their experimental data. A step toward coupling mesoscale microstructure and thermal conductivity has been outlined by Tonks et. al, where fission gas behavior and its impact on thermal conductivity have been considered within MARMOT [45]. The implications of thermal conductivity uncertainty have been reviewed by Pastore *et. al* [46]. This report has explored the uncertainty of fission gas release predictions to thermal conductivity and the uncertainty associated with current BISON models at high burnup and concludes that there is a significant deviation between calculated and measured fission gas release, which causes large discrepancies in predicted thermal conductivity [46]. This would suggest that relating FGR to decreasing conductivity may come with several accuracy issues. All these previous methods to assess thermal conductivity by correlating radiation damage, neglect modeling PD's conductivity degrading mechanisms, which are paramount to understanding extended defect growth, and complex microstructural evolution [22,47-49].

This report investigates the implications of thermal conductivity reduction due to point defects in nuclear fuel. To better emphasize this, only the early-stage microstructure evolution is considered, late-stage phenomena such as fission gas bubbles, dislocation networks, grain subdivision, and other coupled extended defects that have negligible impact on point defect evolution in this damage range are neglected. Consideration of other sink sources and features that degrade conductivity would result in a lower thermal conductivity value than will be seen in this study. In FPCs these effects on conductivity are already incorporated in the expressions as a function of burnup, however, they are not treated as independent processes, but as a collective degradation of the fuel [4]. Through examination of point defects and dislocation loops, the thermal conductivity of UO_2 will be modeled up to 1 dpa of irradiation damage. Our analysis will show how point defects are the most significant contributor to early-stage conductivity degradation. Unlike empirical thermal conductivity models based on post-irradiation examination (PIE) of fuel properties, the current analysis is focusing on in-pile values. There is a concern that PIE data may be skewed because of the annealing that occurs in the time between the fuel being removed and when the fuel is cooled down after decay heating, a process that takes several years. Therefore, we also present an analysis of the effect of thermal recovery in PIE of fuels. Additionally, it will be shown that the single largest reduction of thermal conductivity in UO_2 happens within the timeframe shortly after irradiation. Rate Theory (RT) can be utilized to track the population of point defects in the UO_2 lattice, while the Klemens-Callaway (KC) model can be coupled with RT to find the effect of point defects on lattice conductivity, see Fig. 1. By coupling RT and the KC model for point defects, a predictive, non-empirical model of fuel performance can be developed to simplify, standardize, and improve FPCs for current and potential nuclear fuels.

2. Modeling

Over the damage range of interest, PDs are the primary contributor to degrading conductivity [50]. PD populations are dependent on Frenkel pair production, defect recombination, and absorption due to sinks, which is discussed in Section 2.1. Therefore, we limit extended defect evolution to dislocation loops, considering that point defect concentrations are primarily determined by recombination of interstitial-vacancy pairs. Other sink sources can be modeled in the same manner as loops. KC modeling is coupled with the concentration of defects present in the lattice to determine the lattice conductivity of the fuel, which is presented in Section 2.2. A variable thermal conductivity model to solve the nonlinear heat diffusion equation for defect distributions is described in Section 2.3. Once the link between irradiation conditions and thermal conductivity have been bridged, a variable thermal conductivity model will be solved for various temperature, conductivity, and defect concentration distributions in the fuel rod, which are presented in Sections 3 & 4.1. Finally, Section 4.2 discusses the potential for sources of error between in-pile thermal conductivity and PIE thermal conductivity.

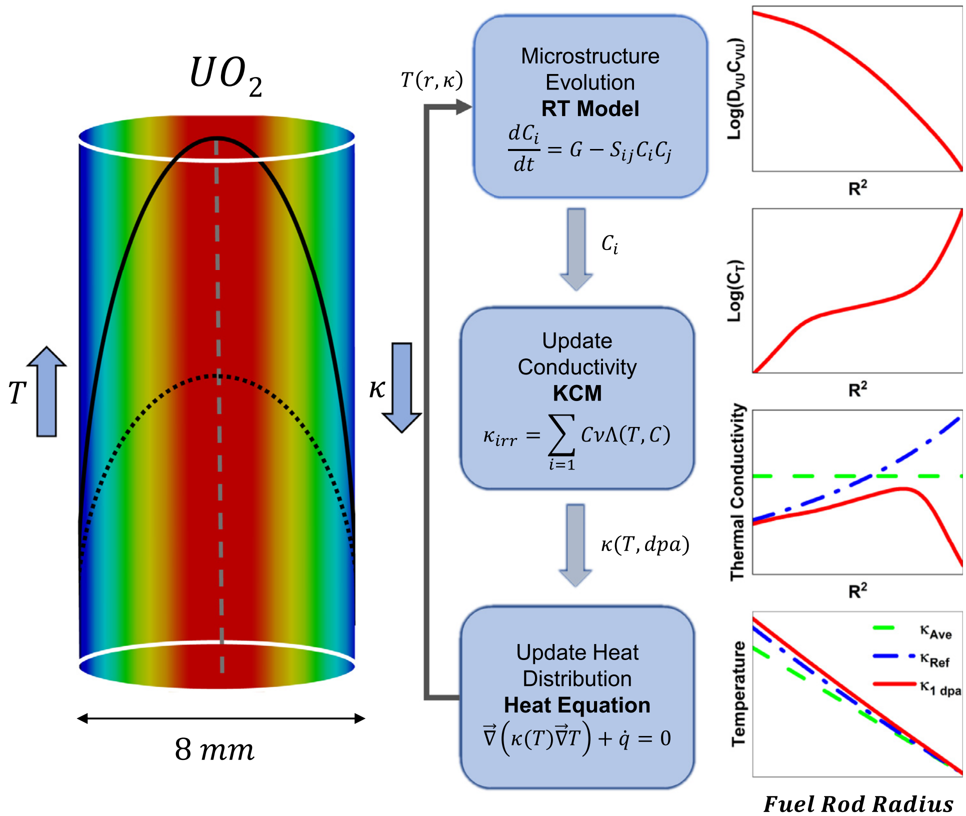

Fig. 1. Outline for the radial temperature calculation procedure.

2.1. Rate Theory

RT allows for the concentrations of defects to be described by a series of coupled differential equations. Defect populations determined by Frenkel pair generation, recombination of self-interstitials and vacancies, and absorption by sinks, such as loops and voids, can be described by Eqs. 1-4 [18,51,52].

$$\frac{dC_{V_U}}{dt} = G_{V_U} - \frac{\Omega_0}{a^2}\left(48D_{U_I} + 48D_{V_U}\right)C_{V_U}C_{U_I} - K_{V_U,S}C_{V_U}C_S \tag{1}$$

$$\frac{dC_{V_O}}{dt} = G_{V_O} - \frac{\Omega_0}{a^2}\left(36D_{O_I} + 24D_{V_O}\right)C_{V_O}C_{O_I} - K_{V_O,S}C_{V_O}C_S \tag{2}$$

$$\frac{dC_{U_I}}{dt} = G_{U_I} - \frac{\Omega_0}{a^2}\left(48D_{U_I} + 48D_{V_U}\right)C_{V_U}C_{U_I} - K_{U_I,S}C_{U_I}C_S \tag{3}$$

$$\frac{dC_{O_I}}{dt} = G_{O_I} - \frac{\Omega_0}{a^2}\left(36D_{O_I} + 24D_{V_O}\right)C_{V_O}C_{O_I} - K_{O_I,S}C_{O_I}C_S \tag{4}$$

These equations describe the concentration of the cation and anion defects of UO_2, namely uranium vacancy V_U, oxygen vacancy V_O, uranium interstitial U_I, and oxygen interstitial O_I. D and C are the diffusion coefficient and the concentration of the respective point defect, while Ω_0 is the volume per atom. The first term of the right-hand side (RHS) of each equation is the defect production rate G, defined in units of dpa/s, and explained further on. The second term on the RHS is the recombination of interstitial and vacancy pairs. The last term is the loss of point defects to sink sources, denoted generally here by subscript s, such as extended defects, grain boundaries, voids, bubbles, and dislocation networks. PD loss due to sink sources includes growth and nucleation of sinks and are summated for each type considered. The corresponding sink rate, K, for common types of sink sources can be found in radiation effects textbooks and summated according to Eq. (5) to find the net sink rate of PDs [23,53]:

$$K_s = K_{\text{dislocations loops}} + K_{\text{loop nucleation}} + K_{\text{voids}} + K_{\text{void nucleation}} + K_{\text{GB}}\ldots \tag{5}$$

PD loss due to dislocation loops is defined as the loss due to sink growth and shrinkage depending on defect type and sink nucleation. For dislocation loops, the sink strength is proportional to the loop density N_L and loop size R_L, and the loop defect species flux j^L [52]. Defect flux of dislocation loops is the equivalent number of defects that enter or leave a toroidal-shaped body and still maintain stoichiometry of the loop. Interstitial defect loss to growing dislocation loops can then be represented as follows [28]:

$$K_{U_I,L}C_{U_I}C_L = \frac{1}{2}K_{O_I,L}C_{O_I}C_L = 2\pi^2 r_0 R_L N_L \times j_i^L = \frac{2\pi^2 r_0 R_L N_L}{r_0 \ln\frac{8R_L}{r_0}} \times \frac{D_{U_I}C_{U_I}D_{O_I}C_{O_I}}{2D_{U_I}C_{U_I} + D_{O_I}C_{O_I}} \tag{6}$$

where a is the lattice constant equal to 0.547 nm, r_0 is the minor radius of the toroid representing the dislocation loop. Loss of vacancy PD's is treated similarly, as vacancy defects combine with dislocation loops to recombine with an interstitial and decrease the size of the loop.

$$K_{V_U,L}C_{V_U}C_L = \frac{1}{2}K_{V_O,L}C_{V_O}C_L = 2\pi^2 r_0 R_L N_L \times j_v^L = \frac{2\pi^2 r_0 R_L N_L}{r_0 \ln\frac{8R_L}{r_0}} \times \frac{D_{V_U}C_{V_U}D_{V_O}C_{V_O}}{2D_{V_U}C_{V_U} + D_{V_O}C_{V_O}} \tag{7}$$

PD loss to dislocation loop nucleation is represented by the product of the defect capture radius $\frac{84\Omega_0}{a^2}$, and the di-interstitial flux. Flux for di-interstitials is governed by the slowest species

Table 1
UO_2 point defect parameters.

Defect Type	Prefactor, $\boldsymbol{D_0}$ (cm^2/s) [60]	Migration barrier, $\boldsymbol{E_m}$ (eV)	Phonon scattering cross-section, $\boldsymbol{S_i^2}$ [61]
V_U	0.65	5.0 [55]	22.6
V_O	0.02	2.1 [54]	7.49
U_I	0.01	2.6 [55]	16.6
O_I	0.01	1.0 [55]	7.68

[52].

$$K_{U_I,L}C_{U_I}C_L = \frac{1}{2}K_{O_I,LN}C_{O_I}C_L = \frac{84\Omega_0}{a^2} \times j_{ii} = \frac{84\Omega_0}{a^2} \times \frac{D_{U_I}C_{U_I}C_{U_I}D_{O_I}C_{O_I}C_{O_I}}{2D_{U_I}C_{U_I}C_{U_I} + D_{O_I}C_{O_I}C_{O_I}} \quad (8)$$

If grain boundaries of size d were to be included as a sink source, the rate constant would be equivalent to the following rate constant in Eq. (9) [25]:

$$K_{GB} = 4\pi D_i d \quad (9)$$

Loop density and loop size are described below by Eqs. (10,11). Dislocation loop growth is evident when the flux of interstitials growing the loop is larger than the flux of vacancies recombining with interstitials and shrinking the loop [28].

$$\frac{dN_L}{dt} = \frac{84\Omega_0}{a^2} j_{ii} \quad (10)$$

$$\frac{dR_L}{dt} = 3\Omega_0 \frac{2\pi r_0}{b}\left(j_i^L - j_v^L\right) - \frac{R_L}{2N_L}\frac{dN_L}{dt} \quad (11)$$

For conversion between the volumetric densities given thus far and dpa, the volumetric concentrations are multiplied by the volume per atom, Ω_0. It was found that the rate constant for recombination is larger than that of loops, which is larger than that of grain boundaries from Eq. (9). This further justifies our inclusion of only recombination and loops in the RT analysis. Diffusion of defects into other areas of fuel according to Fick's law is neglected here. It should be noted that compared to PD loss due to recombination and other sink sources, spatial diffusion plays a minor role. All diffusion coefficients are subject to the temperature dependence of the Arrhenius law, $D = D_o \exp(-\frac{E_m}{k_B T_i})$. D_o is the diffusion coefficient prefactor, and E_m is the migration barrier, shown in Table 1 for each defect type. T_i is referred to as the irradiation temperature. Migration barrier energies of defects were taken from several studies which reported near equal values for all but V_U [54,55]. Migration barrier uncertainties can lead to different populations of defects and alter the total defect scattering cross section utilized below in the KC model.

After 1 dpa, the microstructure can be not described solely by the aforementioned series of RT equations due to not considering other microstructural effects of radiation. Therefore, the damage ranges considered in this analysis will be limited to 1 dpa. Additional formulation of the RT equations to account for various sink sources, in addition to accounting for the interactions between sink sources, can be used to extend the damage threshold past 1 dpa whilst accurately describing the microstructure, but is out of the scope of this study.

Typical nuclear fuel inside a light water reactor experience roughly 1 dpa per day from fission products, yet many of these initially displaced atoms recombine during the damage cascade [26,38]. Rate theory models the effects of PDs after the initial damage cascade recombination. This damage can be best described by the recoil events of the high energy fission products. Assuming the average energy of fission products is on the order of 130 MeV as a primary source of displacement damage, by using SRIM we determine the defect production ratio for oxygen to uranium is 1.34 [56]. This was done by determining the energies of individual fission products, and then simulating the damage ratio of defects from the fission products colliding with UO_2 in SRIM, at the calculated fission energies.

2.2. Klemens-Callaway model

Phonons carry significant thermal energy throughout the atomic lattice of UO_2. Phonon-phonon scattering is due to the anharmonicity of crystal vibrations. At high temperatures, the anharmonicity of phonons reduces the ability of thermal transport in insulators. The Klemens-Callaway model considers phonon-mediated thermal transport in ceramic oxides as proportional to the heat capacity C_s, the phonon velocity ν, and the phonon relaxation time τ_c, integrated over all phonon frequencies ω up to the maximum frequency of the lattice ω_D [33,49,57]. The KC model considers only the contribution of acoustic modes to thermal conductivity and utilizes the Debye approximation of linear phonon dispersion relation and takes the form:

$$k = \int_0^{\omega_D} C_s \nu^2 \tau_c \frac{\omega^2}{2\pi^2\nu^3} d\omega \quad (12)$$

In Eq. (12) The heat capacity of the phonons assumes the high temperature limit, whereby it is equal to the Boltzmann constant, k_B. The upper limit of the integration is known as the Debye frequency (ω_D), and it is the maximum frequency that phonons oscillate at. This upper limit to phonon oscillation is dependent on the phonon group velocity, and the unit cell volume of UO_2. The Debye frequency is given as $\omega_D^3 = \frac{6\pi^2\nu^3}{\Omega_0}$. The total phonon relaxation time τ_c is affected by the intrinsic 3-phonon scattering τ_{3ph}, phonon-impurity scattering τ_i, and phonon-defect scattering τ_{pd}, and phonon-faulted loop scattering τ_L^{-1}. Each scattering mechanism reduces the overall phonon relaxation time, which is calculated according to Matthiessen's Rule as Eq. (13).

$$\tau_c^{-1} = \tau_{3ph}^{-1} + \tau_i^{-1} + \tau_{pd}^{-1} + \tau_L^{-1} \quad (13)$$

The 3-phonon scattering rate $\tau_{3ph}^{-1}$is given by:

$$\tau_{3ph}^{-1} = Be^{-\frac{T_D}{3T_p}}\omega^2 T_p \quad (14)$$

where T_D is the Debye temperature which is taken from literature, and $B = 1.62 \times 10^{-18}$ s/K, which was found by fitting to unirradiated UO_2 conductivity data obtained from [58,59]. T_p is listed here as the physical temperature of the fuel and is different from the irradiation temperature, T_i, which is used to solve for defect concentrations. The impurity scattering rate is a function of the impurity scattering parameter (Γ_i) and was also fitted and N is the number of atoms per unit cell, equal to 6 for UO_2. Scattering rates for impurities and defects are given by:

$$\tau_{i,pd}^{-1} = \frac{\Omega_o}{4\pi N\nu^3}\Gamma_{i,pd}\omega^4 \quad (15)$$

The phonon-defect scattering rate is similar to the impurity scattering parameter, except the point defect parameter (Γ_{pd}) is used in addition to the impurity scattering cross-section. The point defect scattering parameter is proportional to the sum of the concentration of defects times the scattering cross-section of each defect, which can be found in Table 1.

The point defect scattering parameter is given as:

$$\Gamma_{pd} = \sum_i C_i S_i^2 \quad (16)$$

Scattering rates of dislocation loops, voids, and other extended defects and microstructural phenomena can be included in Eq. (14). Less than 1 dpa, these effects on the thermal conductivity are limited when compared to PDs [62]. The scattering rate for stacking

Table 2
Summary of parameters used in the combined model.

Parameter	Value
phonon group velocity (ν) [61]	2.644×10^5 cm/s
Debye frequency (ω_D)	29.9 THz
Debye temperature (T_D) [59]	395 K
volume per atom (Ω_0)	4.094×10^{-23} cm^3
impurity scattering cross-section (Γ_i)	0.077836
three phonon contribution (B)	1.62×10^{-18} s/K

faults is given by:

$$\tau_L^{-1} = \frac{0.7a^2\gamma^2 N_s}{\nu}\omega^2 \tag{17}$$

Where N_s is the number of stacking faults per unit length and equal to $N_s = \pi R_L^2 N_L$. The Gruneisen parameter, γ, is found from [49]. The method for dislocation loop evolution under irradiation and thermal conductivity degradation has been validated previously in [52,63]. All parameters used in the KC model are displayed in Table 2.

Defect scattering cross-sections were taken from [61,62], which were calculated based on molecular dynamics simulations of UO_2 and were converted to scattering cross sections using the procedure outlined in the previous references. As can be seen in Table 1, uranium defects contribute more to decreasing phonon scattering times than oxygen defects. The minimum thermal conductivity is also referred to as the amorphous limit of thermal conductivity, and in UO_2 is assumed to be 1.5 W/mK. To account for this effect, we follow a method similar to one reported in [36], where the maximum scattering rate of a phonon was limited so that each phonon's mean free path, calculated using $\Lambda = \nu\tau_c$, was larger than the lattice constant. This procedure ensures degraded conductivity across all temperatures remains above 1.5 W/mK.

We should distinguish between different temperatures, irradiation temperature T_i, and the temperature at which conductivity is calculated/measured T_p. When both are the same, we refer to this as the fuel temperature (T). Irradiation temperature corresponds to the temperature used to calculate defect mobility and diffusion under irradiation, whereas the measured temperature is the physical temperature of the fuel after irradiation.

2.3. Temperature distribution

The heat conduction equation for the temperature distribution inside the fuel rod is solved using variable thermal conductivity based upon the KC approximation as a function of dose, dose rate, irradiation temperature, and measurement temperature. Between RT and the KC model, the point defect concentration of the fuel rod can be extracted from RT, varying gradually across the radial direction of the fuel rod [64]. This concentration distribution will be used to show where most defects are clustered together inside the fuel rod. The steady state heat diffusion equation is described by Eq. (18).

$$\frac{1}{r}\frac{\partial}{\partial r}\left(\kappa(T)\, r\frac{\partial T_{iso}}{\partial r}\right) = -\dot{q} \tag{18}$$

A variable-step, variable-order differential-algebraic stiff equation solver implemented in MATLAB is used to solve the non-linear differential equation [65]. The parameters used in temperature distribution analysis are shown in Table 3. The outside of the fuel rod maintains a constant surface temperature boundary condition, while the inside of the fuel rod maintains a symmetry boundary condition.

Table 3
Parameters used in heat diffusion equation.

Parameter	Value
Volumetric heat generation rate	600 MW/m^3
Surface Temperature	425°C
Fuel rod radius	4.0 mm

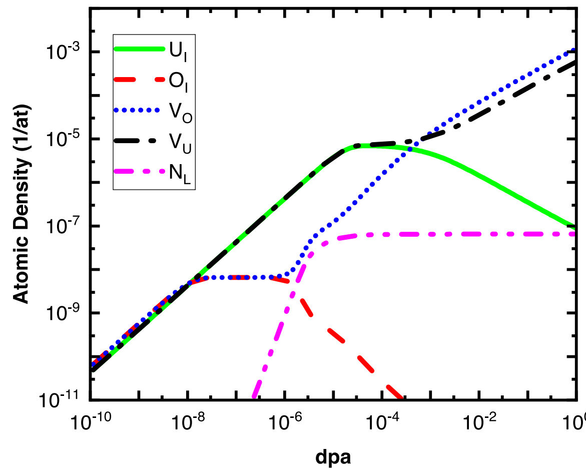

Fig. 2. Defect concentration under irradiation at 1000°C and dose rate of 10^{-6} dpa/s.

3. Results

Fuel temperature in commercial reactors under normal operating conditions varies from 400°C to 1400°C [32]. To demonstrate defect evolution predicted by the RT model we use 1000°C as a reference. In Fig. 2, the concentration of four types of PDs is shown as a function of irradiation dose, by solving Eqs. (1-4). The model suggests that cation defects begin to recombine around 10^{-5} dpa and reach a quasi-equilibrium condition for a short period. Shortly thereafter, interstitial concentrations reduce owing to their higher mobility and absorption by loops. Conversely, the concentration of vacancies increases as they are immobile to undergo any absorption at sinks and there are fewer interstitials to recombine. The same is the case for anion defects, however, due to lower loop density and migration barriers, the recombination process persists longer than that of the cation defects. The anion defects achieve recombination-driven quasi-equilibrium much earlier at 10^{-8} dpa due to their higher mobility and recombination.

O_I concentrations decrease beginning around 10^{-6} dpa due to absorption by loops. This is followed by slower reduction characterized by saturation in loop nucleation. The reduction of U_I at 10^{-4} dpa corresponds to the onset of interstitial loop growth [28]. In general, these results are consistent with RT model predictions for monoatomic systems characterized by different defect evolution stages [53]. A unique aspect of the model for this diatomic system is the independent evolution of cation and anion defects. An interesting feature is that at higher doses, the V_O concentration becomes larger than V_U. Some supporting evidence for this has been discussed in a recent report on defect characterization revealing oxygen vacancies in irradiated ThO_2 using optical spectroscopy [14,66].

Point defect evolution as a function of temperatures at different dose levels is shown in Fig. 3. While the cumulative defect concentration is shown in Fig. 3(a), the individual defect's concentration at 1 dpa is shown in Fig. 3(b). We observe that at temperatures below 600°C cation concentrations are larger, whereas above this temperature concentration of V_O is the largest. This difference can

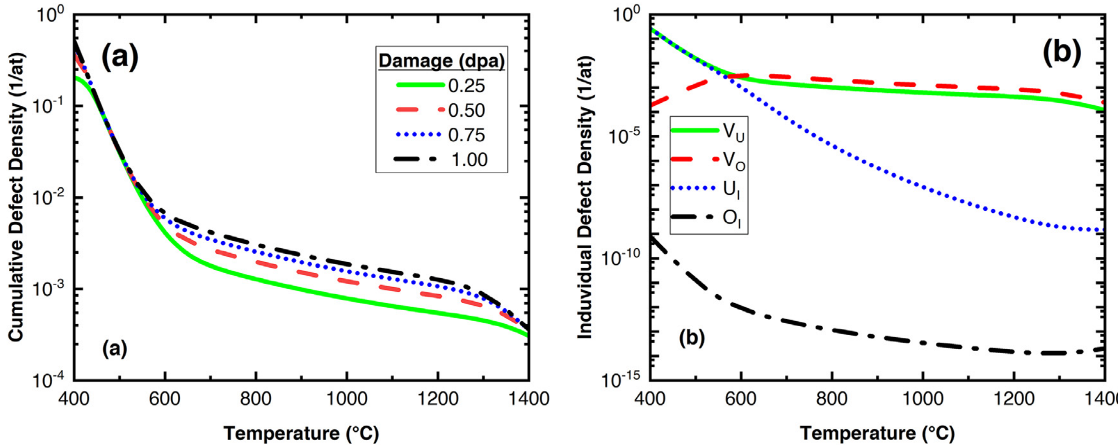

Fig. 3. (a) Cumulative and (b) individual point defect concentration at 1 dpa for various irradiation temperatures.

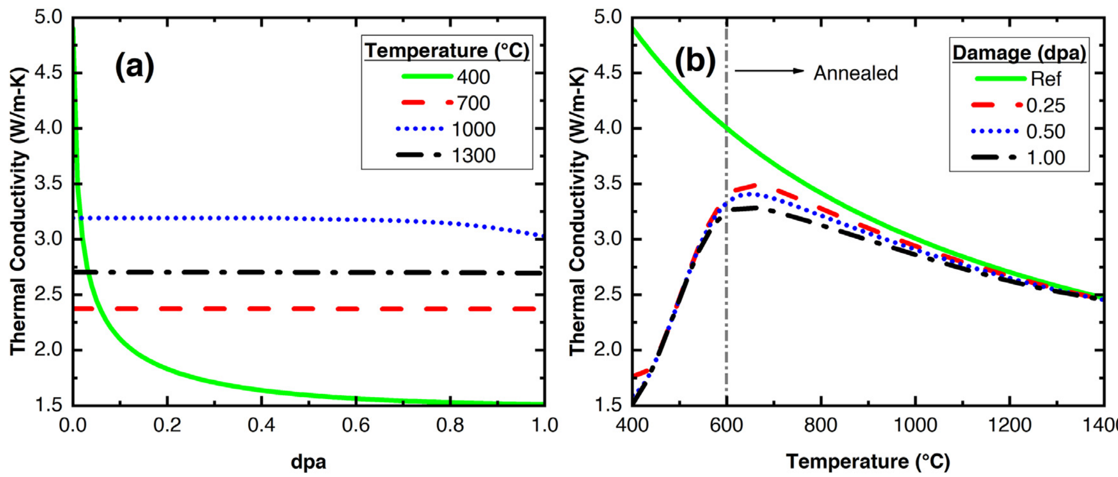

Fig. 4. Thermal conductivity reduction as a function of (a) damage at various fuel temperatures and (b) fuel temperature at various damages. The temperature of stage II annealing is marked with a dashed-dotted line.

be best understood considering different temperature stages used to describe microstructure evolution in irradiated samples [48,67]. The defects produced by irradiation are subject to annealing when the irradiation temperature reaches a certain recovery stage [31]. The reduction of defect concentrations in Fig. 3 corresponds to the thermal recovery stages of defects where defects recombine due to increased mobility, which is dependent on the migration energies. Defect annealing stages have been previously defined, and are commonly seen in three stages, corresponding to three temperature regions [48,67,68]. Stage I recovery occurs at lower temperatures not seen inside nuclear reactors. By 600°C the recovery stage II for UO_2 is characteristic of onset U_I mobility, and gradually decreases the species population due to recombination. Stage III recovery occurs after 1400°C and is often attributed to the onset of V_U mobility and the dissolution of vacancy clusters at high damage.

The reduction of thermal conductivity due to irradiation at various temperatures was found from solving Eqs. (12-17) and is shown in Fig. 4(a). The thermal conductivity of ceramic oxides is known to decrease with temperature, regardless of defects, where the primary mode of phonon interaction is Umklapp scattering [69]. There are two modes present that decrease thermal conductivity. The primary mode is found in both irradiated and unirradiated fuels and arises from the decrease of phonon scattering times as temperature increases [12]. The second mode is due to irradiation-induced defects, as the increase in phonon scattering rates from defects decreases the overall phonon conductivity. This can be seen better in Fig. 4(b), where at low temperatures cation defects have yet to anneal and are found in high concentrations, and the irradiated and unirradiated conductivities diverge due to the latter mode. After defects have finished annealing, the conductivity values seem to converge, indicating only the former mode of conductivity degradation. The large presence of defects below 600°C is the cause of the minimal thermal conductivity seen in Fig. 4(b), which is confirmed in Fig. 4(a) where the conductivity of UO_2 irradiated at 400°C presents a much more damage-dependent conductivity profile.

4. Discussion

4.1. Fuel Rod Distribution

The heat diffusion equation (Eq. (18) can now be solved with the temperature and irradiation damage dependent thermal conductivity model. Paramount to the temperature distribution is the location and diffusion of defect concentrations, which is primarily dependent on temperature and varies along the radial direction of the fuel rod. A feature that is neglected is the increased damage rate because of the higher neutron flux near the rim of the fuel. This would result in higher damage to the rim, and therefore an increase in defect concentrations in that area. This feature has not been explicitly considered in former modeling studies of conductivity in nuclear fuels, as average conductivity, not spatial conduc-

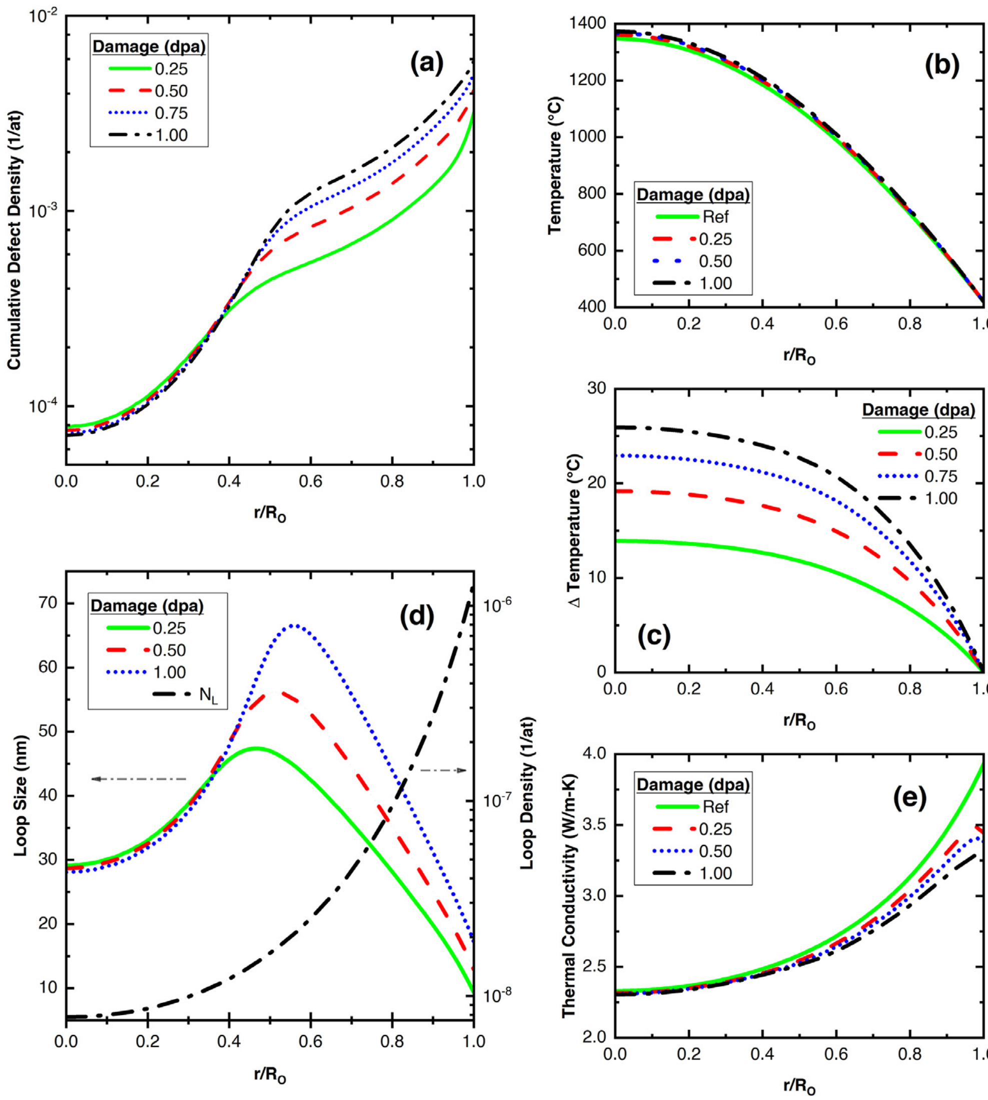

Fig. 5. (a) Cumulative point defect distribution, (b) temperature distribution, (c) temperature difference of irradiated and unirradiated fuel, (d) dislocation loop size and density distribution, (e) and thermal conductivity profile in a UO_2 fuel rod. Decreased conductivity resulting from larger defect immobility cause higher local defect concentrations around the fuel rim. Loop densities do not change between 0.25 – 1 dpa due to reaching saturation.

tivity is sought after [32,44]. In our work, this assumption is still justified because the lack of annealing effects present near the rim will contribute to higher local defect populations. Fig. 5 shows the distribution of various properties inside of a UO_2 fuel rod for the steady state operating conditions of a light water reactor. The comparison of the temperature profile for several different damage scenarios is shown, and an increase in the centerline temperature can be observed in Fig. 5(b) and Fig. 5(c), which describe the temperature distribution, and the difference between unirradiated and irradiated fuel temperature profiles, respectively. Fig. 5(b) shows the temperatures near the rim of the fuel rod are low enough that defects cannot anneal. In turn, the irradiated conductivity decreases due to high defect concentrations as seen in Fig. 5(a) and Fig. 5(e). Because unirradiated UO_2 contains no radiation induced defects, the reference conductivity increases more than the irradiated near the rim of the fuel rod due to Umklapp scattering.

Defects concentrations which are shown in Fig. 5(a) cause an increase in the centerline temperature of roughly 25°C after 1 dpa, with decreasing increments of temperature change at higher damages. Fig. 5(d) presents the size and density distributions of dislocation loops. Only one distribution is shown for loop density in the fuel rod due to there being no variation between different damages greater than 0.25 dpa as it has reached a saturation density. The density distribution of dislocation loops in the UO_2 fuel rod between 0.25 and 1 dpa is entirely dependent on the temperature of the fuel rod. Dislocation loop sizes increase to their maximum near the center of the fuel rod, where temperatures are ideal for dislocation loop growth [20]. The defected microstructure near the edge produces a thermal barrier in the fuel matrix. Large portions of the energy generated in the fuel rod will stay within the fuel, as the reduced conductivity opposes the thermal transport across the rim boundary. This makes the process of cooling the fuel assembly more difficult and consequently contributes to a further rise in the centerline temperature of the fuel [23].

4.2. Defect annealing

RT can be used in the quantification of defect recovery under out-of-pile storage conditions. A common point of contention for fuel conductivity experiments is the recovery of irradiation-induced defects before the thermal conductivity measurements are done as part of PIE [32,44,70,71]. Experiments that have been previously included in FPCs measure the thermal conductivity of nuclear fuels at various temperatures and burnup after they are taken out of dry storage. Throughout the time in wet storage, the fuel re-

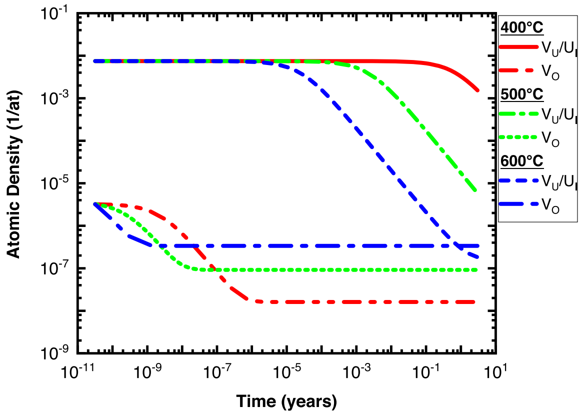

Fig. 6. UO_2 defect recovery evolution under storage conditions at various temperatures after irradiation up to a dose of 0.05 dpa (T_i = 300°C). Both cation defect populations follow the same trends and have the same initial values.

mains radioactive and continues to radiate heat through the form of fission product decay. The large periods spent at high temperatures can anneal the defects present in the sample, causing a difference between the thermal conductivity measured with PIE, and that of the in-pile fuel [70].

We address the potential implication that defect annealing under storage conditions causes different post-irradiation defect concentrations by looking at what temperatures and annealing times produce significant differences in defects. A common safety criterion for wet storage in the U.S. is the fuel cladding temperature should not exceed 400°C, although centerline fuel temperature would be more than this [72]. In wet storage, no neutron irradiation occurs, however, Lucuta et. al. mention the decay of minor actinides does contribute to damaging the fuel during storage [44]. To understand the exclusive effects of thermal recovery of UO_2 during storage, the defect production rate for this demonstration is considered negligible. The same dose rate used previously is considered for UO_2 irradiated at 300°C to a dose of 0.05 dpa to introduce defects. The defect concentration up to 3 years in storage is shown below in Fig. 6 for various annealing temperatures.

The temperature range chosen in Fig. 6 is the earliest that thermal recovery of defects becomes apparent. O_I quickly becomes negligible in concentration and V_O concentrations similarly drop to relatively low densities after 10^{-6} years (0.5 minutes) at 600°C. The varying degrees of time dependent stages II thermal recovery also become more obvious when examined in these temperature ranges. Cation defect concentrations are equal in value, as comparatively few are absorbed by loops.

Below 400°C, PD populations are negligibly affected by thermal recovery. At this temperature, it is unlikely that the fuel temperature will exceed the cladding temperature by much. From this, it can be inferred that in long term storage, the recombination of defects is mostly negligible, as the only defects that are present in considerable concentrations are cation defects, which do not begin to decay until at least temperatures of 500°C are achieved for long periods. Significant cation PD reduction does not occur until high-temperature (>600°C) storage conditions are maintained for roughly 4 hours. The recombination of PD concentrations will cause any conductivity measurements to not be accurately representative of the in-pile microstructure. Considering the safety criterion of 400°C cladding temperature, and that fuel temperature would be more than this, it would be possible for fuel to satisfy the cladding temperature limit, while actively maintaining temperatures that would promote recombination of PDs, altering the PIE measured thermal conductivity from the actual in-pile thermal conductivity. However, the magnitude of this effect is ultimately dependent upon the time, dose, irradiation temperature, and the fuel temperature during storage.

5. Conclusion

A coupled model was presented for determining the thermal conductivity of UO_2 during irradiation by tracking defect concentration generation rates and mobility. This model was not dependent on any post-irradiation examination of nuclear fuels and considers the effects not captured by semi-empirical datasets of UO_2 used by fuel performance codes, such as dose rate and irradiation temperature. A standardized approach for common defect sink sources was introduced to extend the scope to medium to long-range microstructure evolution under irradiation. Defect distributions inside UO_2 fuel rods were shown to indicate an increase in defects near the rim of the fuel rod. The effects of thermal recovery on post-irradiation examination of nuclear fuels were examined to find that only considerable amounts of time at 400°C or short periods at 600°C have the possibility to skew conductivity measurements used in fuel. Rate theory models such as this that require little to no parameterization of data help reduce the time for advanced nuclear fuel qualifications by reducing the number of experiments required to accurately characterize fuel properties.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Joshua Ferrigno: Conceptualization, Investigation, Methodology, Writing – original draft. **Saqeeb Adnan:** Methodology, Writing – review & editing. **Marat Khafizov:** Project administration, Writing – review & editing.

Data Availability

Data will be made available on request.

Acknowledgments

This work was supported by Center for Thermal Energy Transport under Irradiation (TETI), an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, and Office of Basic Energy Sciences.

References

[1] B.C. Cheng, R.L. Yang, Fuel failure experiences in U.S, Light Water Reactors (2003).
[2] H.S. Aybar, P. Ortego, A review of nuclear fuel performance codes, Prog. Nucl. Energy 46 (2) (2005) 127–141.
[3] P.E. MacDonald, L.B. Thompson, in: MATPRO: Version 09. A Handbook Of Materials Properties For Use In The Analysis Of Light Water Reactor Fuel Rod Behavior,; SEE CODE- 9502158 EG and G Idaho, Inc., Idaho Falls (USA), Idaho National Engineering Lab., 1976, p. 402. p. Medium: X; Size: Pages.
[4] P. Van Uffelen, J. Hales, W. Li, G. Rossiter, R. Williamson, A review of fuel performance modelling, J. Nucl. Mater. 516 (2019) 373.
[5] K. Lassmann, TRANSURANUS: a fuel rod analysis code ready for use, J. Nucl. Mater. 188 (1992) 295–302.
[6] R.L. Williamson, J.D. Hales, S.R. Novascone, G. Pastore, K.A. Gamble, B.W. Spencer, W. Jiang, S.A. Pitts, A. Casagranda, D. Schwen, A.X. Zabriskie, A. Toptan, R. Gardner, C. Matthews, W. Liu, H. Chen, BISON: a flexible code for advanced simulation of the performance of multiple nuclear fuel forms, Nucl. Technol. 207 (7) (2021) 954–980.
[7] K. Geelhood, W. Luscher, P. Raynaud, I. Porter, FRAPCON-4.0: a computer code for the calculation of steady-state, Thermal-Mech. Behav. Oxide Fuel Rods High Burnup (2015).

[8] P. Van Uffelen, A. Schubert, J. Van de Laar, C. Gyori, D. Elenkov, S. Boneva, M. Georgieva, S. Georgiev, D. Hozer, C. Martens, G. Spykman, C. Hellwig, L. Nordstrom, L. Luzzi, V. Di Marcello, L. Ott, Verification of the Transuranus fuel performance code - an overview, 7 International Conference on WWER Fuel Performance, Modelling And Experimental Support, 2008.
[9] P.M. Chantoin, E. Sartori, J.A. Turnbull, The compilation of a public domain database on nuclear fuel performance for the purpose of code development and validation, OECD-NEA, Nuclear Energy Agency of the OECD (NEA), 1998.
[10] R. Faibish, S.L. Hayes, C. Stanek, C.A. Back, K. Terrani, Accelerated Fuel Qualification (AFQ) Workshop I Summary Report, 2019.
[11] K.A. Terrani, N.A. Capps, M.J. Kerr, C.A. Back, A.T. Nelson, B.D. Wirth, S.L. Hayes, C.R. Stanek, Accelerating nuclear fuel development and qualification: modeling and simulation integrated with separate-effects testing, J. Nucl. Mater. 539 (2020) 152267.
[12] D.H. Hurley, A. El-Azab, M.S. Bryan, M.W.D. Cooper, C.A. Dennett, K. Gofryk, L. He, M. Khafizov, G.H. Lander, M.E. Manley, J.M. Mann, C.A. Marianetti, K. Rickert, F.A. Selim, M.R. Tonks, J.P. Wharry, Thermal energy transport in oxide nuclear fuel, Chem. Rev. (2021).
[13] D. Gaston, C. Permann, J. Peterson, A. Slaughter, D. Andrš, Y. Wang, M. Short, D. Perez, M. Tonks, J. Ortensi, L. Zou, R. Martineau, Physics-based multiscale coupling for full core nuclear reactor simulation, Ann. Nucl. Energy 84 (2015) 45–54.
[14] C.A. Dennett, W.R. Deskins, M. Khafizov, Z. Hua, A. Khanolkar, K. Bawane, L. Fu, J.M. Mann, C.A. Marianetti, L. He, D.H. Hurley, A. El-Azab, An integrated experimental and computational investigation of defect and microstructural effects on thermal transport in thorium dioxide, Acta Mater. 213 (2021) 116934.
[15] M. Bertolus, M. Freyss, B. Dorado, G. Martin, K. Hoang, S. Maillard, R. Skorek, P. Garcia, C. Valot, A. Chartier, L. Van Brutzel, P. Fossati, R.W. Grimes, D.C. Parfitt, C.L. Bishop, S.T. Murphy, M.J.D. Rushton, D. Staicu, E. Yakub, S. Nichenko, M. Krack, F. Devynck, R. Ngayam-Happy, K. Govers, C.S. Deo, R.K. Behera, Linking atomic and mesoscopic scales for the modelling of the transport properties of uranium dioxide under irradiation, J. Nucl. Mater. 462 (2015) 475–495.
[16] X.Y. Liu, C. Matthews, W. Jiang, M.W.D. Cooper, J.D. Hales, D.A. Andersson, Atomistic and cluster dynamics modeling of fission gas (Xe) diffusivity in TRISO fuel kernels, J. Nucl. Mater. 561 (2022) 153539.
[17] Impact of High Burnup Uranium Oxide and Mixed Uranium-Plutonium Oxide Water Reactor Fuel on Spent Fuel Management, 2011.
[18] A.T. Motta, D.R. Olander, B. Wirth, Light Water Reactor Materials, American Nuclear Society, 2017.
[19] A. Toptan, R.K. Salko, M.N. Avramova, K. Clarno, D.J. Kropaczek, A new fuel modeling capability, CTFFuel, with a case study on the fuel thermal conductivity degradation, Nucl. Eng. Des. 341 (2019) 248–258.
[20] G.S. Was, Fundamentals of radiation materials science: metals and alloys, Second edition ed.
[21] B. Szpunar, D. Oladimeji, J. Ranasinghe, I. Rossland, L. Malakkal, E. Jossou, J.A. Szpunar, First principles studies of thermal conductivity of nuclear fuel materials, CANDU Fuel: Evolution Towards Optimal Performance 13th International Conference on CANDU fuel, Canadian Nuclear Society, Canada, 2016 666Megabytes.
[22] T. Wiss, 2.18 - radiation effects in UO2, in: R.J.M. Konings (Ed.), Comprehensive Nuclear Materials, Elsevier, Oxford, 2012, pp. 465–480.
[23] D.R. Olander, Fundamental aspects of nuclear reactor fuel elements, in: p. Medium: ED; Size: Pages, California Univ., Berkeley (USA). Dept. of Nuclear Engineering, 1976, p. 624.
[24] J.R. Lamarsh, A.J. Baratta, Introduction to Nuclear Engineering, Prentice Hall2011.
[25] S.I. Golubov, A.V. Barashev, R.E. Stoller, 1.13 - radiation damage theory, in: J.M.K. Rudy (Ed.), Comprehensive Nuclear Materials, Elsevier, Oxford, 2012, pp. 357–391.
[26] S.J. Zinkle, Radiation-induced effects on microstructure, in: R.J.M. Konings (Ed.), Comprehensive Nuclear Materials, Elsevier, Oxford, 2012, pp. 65–98.
[27] K. Chiken Kinoshita, Kazunori Hayashi, Kinetics of point defects in electron irradiated MgO, Nucl. Instrum. Methods Phys. Res. Sect. B 1 (2) (1984) 209–218.
[28] S.C. Vinay, J. Pakarinen, T. Yao, L. He, D.H. Hurley, M. Khafizov, Indirect characterization of point defects in proton irradiated ceria, Materialia 15 (2021) 101019.
[29] S. Zinkle, C. Kinoshita, Defect production in ceramics, J. Nucl. Mater. 251 (1997) 200–217.
[30] A. Resnick, K. Mitchell, J. Park, E. Farfán, T. Yee, Thermal transport study in actinide oxides with point defects, Nuclear Eng. Technol. 51 (5) (2019) 1398–1405.
[31] R.O. Lokken, E.L. Courtright, Review of the effects of burnup on the thermal conductivity of UO2, United States, 1976, p. 15.
[32] C. Ronchi, M. Sheindlin, D. Staicu, M. Kinoshita, Effect of burn-up on the thermal conductivity of uranium dioxide up to 100000 MWd t-1, J. Nucl. Mater. 327 (1) (2004) 58–76.
[33] P.G. Klemens, Thermal conductivity and lattice vibration modes, Solid State Phys.-Adv. Res. Appl. 7 (1958) 1–98.
[34] P.G. Klemens, The scattering of low-frequency lattice waves by static imperfections, Proc. Phys. Soc. London Section A 68 (12) (1955) 1113–1128.
[35] B.-H. Lee, Y.-H. Koo, D.-S. Sohn, Rim characteristics and their effects on the thermal conductivity in high burnup UO2 fuel, J. Nucl. Sci. Technol. 38 (1) (2001) 45–52.
[36] D.G. Cahill, S.K. Watson, R.O. Pohl, Lower limit to the thermal conductivity of disordered crystals, Phys. Rev. B 46 (10) (1992) 6131–6140.
[37] L.O. Jernkvist, A. Massih, Analysis of the Effect of UO2 High Burnup Microstructure on Fission Gas Release, (2002).
[38] V.V. Rondinella, T. Wiss, The high burn-up structure in nuclear fuel, Mater. Today 13 (12) (2010) 24–32.
[39] K. Lassmann, C. O'Carroll, J.v.d. Laar, C.T. Walker, The radial distribution of plutonium in high burnup UO2 fuels, J. Nucl. Mater. 208 (3) (1994) 223–231.
[40] J. Spino, D. Papaioannou, Lattice parameter changes associated with the rim-structure formation in high burn-up UO2 fuels by micro X-ray diffraction, J. Nucl. Mater. 281 (2) (2000) 146–162.
[41] T. Pavlov, F. Kremer, R. Dubourg, A. Schubert, P. Van Uffelen, Towards a More Detailed Mesoscale Fission Product Analysis in Fuel Performance Codes: A Coupling of the Transuranus and MFPR-F Codes, TopFuel, Prague, Czech Republic, 2018.
[42] M. Abdoelatef, F. Badry, K. Ahmed, S. Biswas, A. Jokisaari, D. Schwen, Y. Zhang, C. Permann, Mesoscale Modeling of High Burn-up Structure (HBS) Formation and Evolution in Metallic Fuels, 2019.
[43] F. Cappia, D. Pizzocri, A. Schubert, P. Van Uffelen, G. Paperini, D. Pellottiero, R. Macián-Juan, V.V. Rondinella, Critical assessment of the pore size distribution in the rim region of high burnup UO2 fuels, J. Nucl. Mater. 480 (2016) 138–149.
[44] P.G. Lucuta, H. Matzke, I.J. Hastings, A pragmatic approach to modelling thermal conductivity of irradiated UO2 fuel: Review and recommendations, J. Nucl. Mater. 232 (2) (1996) 166–180.
[45] M.R. Tonks, P.C. Millett, P. Nerikar, S. Du, D. Andersson, C.R. Stanek, D. Gaston, D. Andrs, R. Williamson, Multiscale development of a fission gas thermal conductivity model: coupling atomic, meso and continuum level simulations, J. Nucl. Mater. 440 (1) (2013) 193–200.
[46] G. Pastore, L.P. Swiler, J. Hales, S. Novascone, D.M. Perez, B. Spencer, L. Luzzi, P. Van Uffelen, R. Williamson, Uncertainty and sensitivity analysis of fission gas behavior in engineering-scale fuel modeling, J. Nucl. Mater. 456 (2015) 398–408.
[47] T. Sonoda, M. Kinoshita, I.L.F. Ray, T. Wiss, H. Thiele, D. Pellottiero, V.V. Rondinella, H. Matzke, Transmission electron microscopy observation on irradiation-induced microstructural evolution in high burn-up UO2 disk fuel, Nucl. Instrum. Methods Phys. Res. Sect. B 191 (1) (2002) 622–628.
[48] S.J. Zinkle, 1.03 - Radiation-Induced Effects on Microstructure**Prepared for the Oak Ridge National Laboratory under Contract No. DE-AC05-000R22725, in: R.J.M. Konings (Ed.), Comprehensive Nuclear Materials, Elsevier, Oxford, 2012, pp. 65–98.
[49] J. Pakarinen, M. Khafizov, L. He, C. Wetteland, J. Gan, T.N. Andrew, D.H. Hurley, A. El-Azab, T.R. Allen, Microstructure changes and thermal conductivity reduction in UO2 following 3.9MeV He2+ ion irradiation, J. Nucl. Mater. 454 (1) (2014) 283–289.
[50] M. Jin, C.A. Dennett, D.H. Hurley, M. Khafizov, Impact of small defects and dislocation loops on phonon scattering and thermal transport in ThO2, J. Nucl. Mater. 566 (2022) 153758.
[51] V.S. Chauhan, J. Pakarinen, T. Yao, L. He, D.H. Hurley, M. Khafizov, Indirect characterization of point defects in proton irradiated ceria, Materialia 15 (2021) 101019.
[52] L. He, M. Khafizov, C. Jiang, B. Tyburska-Püschel, B.J. Jaques, P. Xiu, P. Xu, M.K. Meyer, K. Sridharan, D.P. Butt, J. Gan, Phase and defect evolution in uranium-nitrogen-oxygen system under irradiation, Acta Mater. 208 (2021) 116778.
[53] G.S. Was, Ohio Library and Information Network., Fundamentals of radiation materials science: metals and alloys, 2nd ed.
[54] B. Liu, D.S. Aidhy, Y. Zhang, W.J. Weber, Theoretical investigation of thermodynamic stability and mobility of the oxygen vacancy in ThO2–UO2 solid solutions, Phys. Chem. Chem. Phys. 16 (46) (2014) 25461–25467.
[55] M. Bertolus, M. Freyss, M. Krack, R. Devanathan, Assessment of current atomic scale modelling methods for the investigation of nuclear fuels under irradiation: example of uranium dioxide, in: Nuclear Energy Agency of the OECD (NEA), 2015, pp. 248–278.
[56] J.F. Ziegler, J.P. Biersack, M.D. Ziegler, SRIM, the Stopping and Range of Ions in Matter, SRIM Co., Chester, Maryland, 2008.
[57] M. Khafizov, P. In-Wook, C. Aleksandr, H. Lingfeng, L. Jianliang, J. M.J, S. David, L. Thomas, R. P.S, H. E.-A.H.D, Thermal conductivity in nanocrystalline ceria thin films, J. Am. Ceram. Soc. 97 (2) (2014) 562–569.
[58] IAEA, Thermophysical Properties of Materials for Nuclear Engineering: A Tutorial and Collection of Data, 2009.
[59] G. Dolling, R. Cowley, A. Woods, The Crystal Dynamics of Uranium Dioxide, Can. J. Phys. 43 (1965) 1397–1413.
[60] S.P. Waldow, R.A. De Souza, Is excess faster than deficient? A molecular-dynamics study of oxygen-interstitial and oxygen-vacancy diffusion in CeO2, J Phys Condens Matter 2 (2020) 024001.
[61] M. Khafizov, M.F. Riyad, Y. Wang, J. Pakarinen, L. He, T. Yao, A. El-Azab, D. Hurley, Combining mesoscale thermal transport and x-ray diffraction measurements to characterize early-stage evolution of irradiation-induced defects in ceramics, Acta Mater. 193 (2020) 61–70.
[62] M. Jin, Impact of small defects and dislocation loops on phonon scattering and thermal transport in ThO2, Submitted J. Nuclear Mater. (2021).
[63] L. He, T. Yao, K. Bawane, M. Jin, C. Jiang, X. Liu, W.-Y. Chen, J.M. Mann, D.H. Hurley, J. Gan, M. Khafizov, Dislocation loop evolution in Kr-irradiated ThO2, J. Am. Ceram. Soc. 105 (8) (2022) 5419–5435.
[64] N.E. Todreas, M.S. Kazimi, Nuclear Systems, Taylor & Francis Group, New York, 1990.
[65] L. Shampine, M. Reichelt, The MATLAB ODE Suite, SIAM J. Sci. Comput. (1997) 1–22.

[66] A. Khanolkar, C.A. Dennett, Z. Hua, J.M. Mann, D.H. Hurley, M. Khafizov, Inferring relative dose-dependent color center populations in proton irradiated thoria single crystals using optical spectroscopy, Phys. Chem. Chem. Phys. 24 (10) (2022) 6133–6145.
[67] W.J. Weber, Alpha-irradiation damage in CeO2, UO2 and PuO2, Radiat. Eff. 83 (1-2) (1984) 145–156.
[68] J.T. Graham, Y. Zhang, W.J. Weber, Irradiation-induced defect formation and damage accumulation in single crystal CeO2, J. Nucl. Mater. 498 (2018) 400–408.
[69] P.G. Klemens, Thermal conductivity of solids at low temperatures, in: S. Flügge (Ed.), Low Temperature Physics I /Kältephysik I, Springer Berlin Heidelberg, Berlin, Heidelberg, 1956, pp. 198–281.
[70] R.J. White, The growth of intra-granular bubbles in post-irradiation annealed UO2 fuel, Int. Atomic Energy Agency (IAEA) (2001) 91–104.
[71] D. Staicu, V.V. Rondinella, C.T. Walker, D. Papaioannou, R.J.M. Konings, C. Ronchi, M. Sheindlin, A. Sasahara, T. Sonoda, M. Kinoshita, Effect of burn-up on the thermal conductivity of uranium–gadolinium dioxide up to 100GWd/tHM, J. Nucl. Mater. 453 (1) (2014) 259–268.
[72] E.W. Brach, Spent Fuel Project Office Interim Staff Guidance - 11, Revision 3, 2003.